SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Delhi-NCR Campus, Modinagar, Ghaziabad (U.P.)

ASSIGNMENT - IV

(B. Tech. III -Year, VI-SEMESTER)

SUBJECT NAME: Data Science
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Submitted To:

Ms. Geetanjali Tyagi
(Assistant Professor)
Dept. of Computer Science & Technology.
SRM-IST, Delhi-NCR Campus
Modinagar, (Ghaziabad)

Submitted By:

Name: Manomay Dutt Bisht
Reg No: RA2311030030031
Branch: CSE-CYBER SECURITY
Section: "A"

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Explain the concept of data wrangling & its importance.

Ans It is a process of organizing, cleaning and transforming raw data into a structured format so it can be used for analysis. In real world scenarios, raw data collected from different sources is often incomplete, inconsistence or unorganized.

Importance of Data wrangling

1. Improve the quality and reliability of data.
2. Helps in removing errors, inconsistent and duplicates.
3. Makes data structure and ready for analysis.
4. Saves time during analysis and modeling.

2. Explain string manipulation in python with suitable code examples.

Ans String manipulation refers to the process of modifying, analysing or processing text data using various string function available in python. Since many datasets contain text informat^n such as names, addresses or messages, strings manipulation is an important part of data processing. Python provids many building methods for working with strings such as changing case, searching, replacing, slicing and concatenation.

Codes

```
text = "Hello World"
print (text.lower())
print (text.upper())
print (text.replace("World", "Python"))
print (text[0:5])
name = "Yogesh"
greeting = "Hello" + name
print (greeting)
```

String manipulation helps in cleaning and processing textual data before analysis.

(3) Explain the general techniques and programming practices used for efficient handling of large datasets in data science.

Ans Handling large dataset efficiently is an important challenge in data Science because large amount of data require more memory, processing, processing power and time. Therefore, several techniques and practices are used to process large datasets effectively.

Common techniques:

(1) Data Sampling: Analyze a small representative portion of data.

(2) Chunk Processing: Read data in small parts instead of location of entire dataset

(3) Using Efficient Libraries: Tools like Numpy and Pandas for faster processing.

④ Parallel Processing: Use multiple CPU cores to process data faster.

⑤ Data Compression: Reduce storage and memory usage.

④ Explain different data transformat^n techniques used in preprocessing.

Ans. Data transformat^n is a preprocessing step where raw data is converted into a suitable format for analysis or machine learning algorithm. This step ensure that the data is consistent, standardized and meaning full.

Transformation techniques include:-

① Normalization: scaling values b/w 0 & 1

② Standardization: Adjusting data to have mean 0 & standard deviation 1.

③ Encoding: Converting categorical data into numerical form.

④ Aggregation: Combination multiple values into summary statics.

⑤ Smoothing: Removing noise from data.